

CIL MT — SYSTEM (Post Code 14) | Paper-II Practice Sheets

Pattern followed: Paper-II — Professional Knowledge (CS/IT), 100 MCQs × 1 mark, no negative marking, bilingual, 3 hrs for both papers combined (Advt. No. 03/2026 official syllabus).

Sheet design: 10 sheets × 25 questions = 250 questions. Each sheet is a scaled-down full paper, mixed across all ten notified sections in exam proportion:

Section	Q per sheet
Engineering Mathematics	3
Digital Logic	2
Computer Organization & Architecture	3
Programming & Data Structures	3
Algorithms	3
Theory of Computation	2
Compiler Design	2
Operating System	3
Databases	2
Computer Networks	2

Suggested timing: 25 questions in 22 minutes per sheet. Attempt every question — there is no negative marking.

Answer keys for all 10 sheets are in a single section at the very end of this document.

SHEET 1

- Q1.** The number of reflexive relations that can be defined on a set of 4 elements is
(A) 2^{10} (B) 2^{16} (C) 2^4 (D) 2^{12}
- Q2.** The eigenvalues of the matrix $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ are
(A) 2 and 2 (B) 1 and 3 (C) 0 and 4 (D) 1 and 2
- Q3.** A and B are independent events with $P(A) = 0.4$ and $P(B) = 0.5$. Then $P(A \cup B)$ is
(A) 0.60 (B) 0.20 (C) 0.70 (D) 0.90
- Q4.** The minimum number of 2-input NAND gates required to implement a 2-input XOR function is
(A) 3 (B) 6 (C) 5 (D) 4
- Q5.** The range of integers representable in 8-bit 2's complement form is
(A) -128 to $+127$ (B) 0 to 255 (C) -128 to $+128$ (D) -127 to $+127$
- Q6.** A 5-stage pipeline has a stage delay of 2 ns and no hazards. The time to execute 100 instructions is
(A) 1000 ns (B) 210 ns (C) 208 ns (D) 200 ns
- Q7.** The addressing mode most suitable for accessing successive elements of an array inside a loop is
(A) Register (B) Indexed (C) Immediate (D) Direct
- Q8.** The number of address lines required for a 4 KB byte-addressable memory is
(A) 16 (B) 11 (C) 10 (D) 12
- Q9.** The number of distinct binary search trees that can be formed with 3 distinct keys is
(A) 3 (B) 9 (C) 6 (D) 5
- Q10.** The postfix form of the expression $(A + B) * (C - D)$ is
(A) $AB+CD-*$ (B) $+AB-CD*$ (C) $ABCD+-*$ (D) $AB+C-D*$
- Q11.** The minimum number of stacks required to implement a queue is
(A) 3 (B) 2 (C) 1 (D) 4
- Q12.** The solution of $T(n) = 2T(n/2) + n$, $T(1) = 1$ is
(A) $\Theta(n^2)$ (B) $\Theta(n \log n)$ (C) $\Theta(n)$ (D) $\Theta(\log n)$
- Q13.** Quicksort is applied to an already sorted array of n elements with the first element chosen as pivot. The running time is
(A) $O(\log n)$ (B) $O(n)$ (C) $O(n^2)$ (D) $O(n \log n)$
- Q14.** The minimum number of comparisons required to find both the minimum and the maximum of n elements is
(A) $\lceil 3n/2 \rceil - 2$ (B) $n - 1$ (C) $2n - 2$ (D) $n \log n$
- Q15.** The language $L = \{a^n b^n \mid n \geq 1\}$ is
(A) regular (B) finite (C) context-free but not regular (D) not context-free
- Q16.** The minimum number of states in a DFA over $\{a, b\}$ that accepts all strings in which the number of a 's is divisible by 3 is
(A) 4 (B) 3 (C) 6 (D) 2
- Q17.** Assigning a real value to an integer variable without a cast is detected during
(A) syntax analysis (B) code generation (C) lexical analysis (D) semantic analysis
- Q18.** Elimination of left recursion from a grammar is essential for
(A) code optimization (B) LR parsing (C) top-down parsing (D) lexical analysis
- Q19.** A process executes the statements `fork(); fork();`. The total number of processes in existence (including the original) is
(A) 4 (B) 2 (C) 8 (D) 3
- Q20.** Belady's anomaly is exhibited by which page replacement policy?
(A) FIFO (B) LRU (C) LFU (D) Optimal
- Q21.** Which of the following is NOT one of the four necessary conditions for deadlock?
(A) Priority inversion (B) Circular wait (C) Hold and wait (D) Mutual exclusion
- Q22.** $R(A, B, C, D)$ has the FD set $\{AB \rightarrow C, C \rightarrow D, D \rightarrow A\}$. The highest normal form satisfied by R is
(A) 2NF (B) 1NF (C) 3NF (D) BCNF
- Q23.** In a B+ tree index, pointers to actual data records are stored in
(A) both internal and leaf nodes (B) leaf nodes only (C) the root only (D) internal nodes only
- Q24.** The number of usable host addresses in the subnet 192.168.10.0/26 is
(A) 126 (B) 64 (C) 30 (D) 62
- Q25.** Which of the following normally uses UDP at the transport layer?
(A) DNS query (B) FTP control connection (C) SMTP (D) HTTP
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SHEET 2

- Q1.** The number of edges in the complete graph K_7 is
(A) 14 (B) 42 (C) 21 (D) 49
- Q2.** The rank of the matrix $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ is
(A) 1 (B) 0 (C) 2 (D) not defined
- Q3.** A fair coin is tossed 4 times. The probability of getting exactly 2 heads is
(A) $1/4$ (B) $1/2$ (C) $3/8$ (D) $5/16$
- Q4.** The Boolean expression $A'B + AB$ simplifies to
(A) B (B) $A + B$ (C) A (D) AB
- Q5.** The minimum number of flip-flops required to design a MOD-12 counter is
(A) 12 (B) 4 (C) 3 (D) 6
- Q6.** A pipeline stall caused by a conditional branch instruction is an example of
(A) no hazard (B) control hazard (C) structural hazard (D) data hazard
- Q7.** A memory chip organized as $1K \times 8$ requires
(A) 1024 address lines, 8 data lines (B) 10 address lines, 1 data line (C) 8 address lines, 10 data lines (D) 10 address lines, 8 data lines
- Q8.** Which I/O technique requires the least CPU involvement per word transferred?
(A) DMA (B) Programmed I/O (C) Interrupt-driven I/O (D) Polling
- Q9.** Which data structure is used for the evaluation of a postfix expression?
(A) Circular queue (B) Stack (C) Priority queue (D) Queue
- Q10.** The worst-case time to insert a node at the beginning of a singly linked list with n nodes is
(A) $O(\log n)$ (B) $O(n)$ (C) $O(1)$ (D) $O(n \log n)$
- Q11.** The height of a complete binary tree with n nodes (root at height 0) is
(A) $\log_2(n + 1)$ (B) $n/2$ (C) $\lceil \log_2 n \rceil$ (D) $\lceil n/2 \rceil$
- Q12.** The maximum number of comparisons made by binary search on a sorted array of 1000 elements is
(A) 999 (B) 10 (C) 500 (D) 100
- Q13.** Kruskal's algorithm for minimum spanning tree is based on which design technique?
(A) Greedy (B) Divide and conquer (C) Dynamic programming (D) Backtracking
- Q14.** Dijkstra's single-source shortest path algorithm may produce incorrect results when the graph has
(A) more than one source (B) parallel edges (C) cycles (D) negative edge weights
- Q15.** The minimum number of states in a DFA accepting the language of the regular expression $(a + b)^*abb$ is
(A) 5 (B) 3 (C) 6 (D) 4
- Q16.** The pumping lemma for regular languages is used to prove that a given language is
(A) decidable (B) context-free (C) not regular (D) regular
- Q17.** The output of the lexical analysis phase is
(A) a stream of tokens (B) a symbol table only (C) a parse tree (D) intermediate code
- Q18.** Which parser is the most powerful among the following?
(A) LL(1) (B) SLR(1) (C) CLR(1) (D) LALR(1)
- Q19.** Which CPU scheduling algorithm is preemptive by design and gives every process a fixed time slice?
(A) SJF (B) Round Robin (C) Priority (non-preemptive) (D) FCFS
- Q20.** Thrashing in a virtual memory system is directly caused by
(A) contiguous allocation (B) too few processes in memory (C) large page size (D) an excessively high page fault rate
- Q21.** Which of the following is NOT a requirement of a correct critical-section solution?
(A) Mutual exclusion (B) Progress (C) Assumptions about relative process speeds (D) Bounded waiting
- Q22.** Which SQL set operator eliminates duplicate rows by default?
(A) JOIN (B) CROSS JOIN (C) UNION (D) UNION ALL
- Q23.** $R(A, B, C, D)$ has A as its only candidate key. The number of superkeys of R is
(A) 4 (B) 8 (C) 16 (D) 15
- Q24.** In Go-Back-N ARQ using 3-bit sequence numbers, the maximum sender window size is
(A) 4 (B) 7 (C) 8 (D) 3
- Q25.** The ping utility works using which protocol?
(A) ICMP (B) ARP (C) IGMP (D) SNMP
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SHEET 3

- Q1.** The number of onto (surjective) functions from a set of 4 elements to a set of 3 elements is
(A) 24 (B) 36 (C) 64 (D) 81
- Q2.** The determinant of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$ is
(A) 0 (B) 24 (C) 1 (D) -6
- Q3.** For a binomial distribution with $n = 10$ and $p = 0.4$, the mean is
(A) 4 (B) 2.4 (C) 0.4 (D) 10
- Q4.** The hexadecimal equivalent of the binary number 11011010 is
(A) D8 (B) AD (C) DA (D) BD
- Q5.** A full adder can be realized using
(A) two half adders only (B) three half adders (C) one half adder and one AND gate (D) two half adders and one OR gate
- Q6.** In a direct-mapped cache with 1024 blocks and a block size of 8 bytes, the number of bits in the block-offset field of the address is
(A) 8 (B) 3 (C) 2 (D) 10
- Q7.** The register that always holds the address of the next instruction to be fetched is the
(A) Program Counter (B) MAR (C) Instruction Register (D) Accumulator
- Q8.** Zero-address instructions are used in machines organized around
(A) memory-to-memory operations (B) general-purpose registers (C) a stack (D) an accumulator
- Q9.** In C, for the declaration `int a[5];` the expression `a[2]` is equivalent to
(A) `a + 2` (B) `*(a + 2)` (C) `&a + 2` (D) `*a + 2`
- Q10.** The in-order traversal of a binary search tree yields the keys in
(A) level order (B) random order (C) reverse sorted order (D) sorted ascending order
- Q11.** The maximum number of nodes in a binary tree of height h (root at height 0) is
(A) $2h + 1$ (B) $2^h - 1$ (C) $2^{(h+1)} - 1$ (D) 2^h
- Q12.** The auxiliary space complexity of the standard merge sort on arrays is
(A) $O(n \log n)$ (B) $O(n)$ (C) $O(1)$ (D) $O(\log n)$
- Q13.** Which of the following sorting algorithms is stable?
(A) Quicksort (B) Heapsort (C) Merge sort (D) Selection sort
- Q14.** In a hash table with n keys, m slots and collision resolution by separate chaining, the average time for an unsuccessful search is
(A) $O(n)$ (B) $O(1)$ (C) $O(1 + n/m)$ (D) $O(\log n)$
- Q15.** Which class of languages is NOT closed under complementation?
(A) Context-free languages (B) Finite languages (C) Recursive languages (D) Regular languages
- Q16.** The minimum number of states in a DFA over $\{0, 1\}$ accepting all strings ending with 01 is
(A) 2 (B) 3 (C) 5 (D) 4
- Q17.** An S-attributed syntax-directed definition uses
(A) no attributes (B) both, evaluated left to right (C) only synthesized attributes (D) only inherited attributes
- Q18.** The minimum number of temporary variables needed in the three-address code for $a = b * c + d * e$ is
(A) 3 (B) 2 (C) 4 (D) 1
- Q19.** The Banker's algorithm is used for
(A) deadlock avoidance (B) deadlock recovery (C) deadlock detection (D) page replacement
- Q20.** In a system with a 32-bit logical address space and 4 KB pages, the number of pages in the logical address space is
(A) 2^{12} (B) 2^{10} (C) 2^{20} (D) 2^{32}
- Q21.** Which of the following is NOT an inter-process communication mechanism?
(A) Message passing (B) Pipes (C) Shared memory (D) Paging
- Q22.** Which of the following is NOT a primitive (basic) operation of relational algebra?
(A) Natural join (B) Selection (C) Projection (D) Union
- Q23.** Two-phase locking (2PL) protocol guarantees
(A) no starvation (B) recoverability only (C) deadlock freedom (D) conflict serializability
- Q24.** The length of an IPv6 address is
(A) 64 bits (B) 32 bits (C) 256 bits (D) 128 bits
- Q25.** A LAN switch operates primarily at which OSI layer?
(A) Transport (B) Physical (C) Network (D) Data link
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SHEET 4

- Q1.** The number of distinct arrangements of the letters of the word "LEVEL" is
(A) 120 (B) 20 (C) 60 (D) 30
- Q2.** If A is a 3×3 matrix with $\det(A) = 2$, then $\det(2A)$ is
(A) 4 (B) 16 (C) 32 (D) 8
- Q3.** For a Poisson distribution
(A) mean = variance (B) mean < variance (C) mean > variance (D) variance = mean²
- Q4.** The 1's complement of the binary number 0101 is
(A) 0110 (B) 1010 (C) 1011 (D) 1101
- Q5.** The number of select (control) lines required for a 4-to-1 multiplexer is
(A) 2 (B) 4 (C) 1 (D) 3
- Q6.** In a little-endian machine, the least significant byte of a multi-byte word is stored at
(A) the highest address (B) a separate segment (C) the lowest address (D) the stack top
- Q7.** The cache mapping scheme in which a memory block may be placed in any cache line is
(A) sector mapped (B) fully associative (C) direct mapped (D) set associative
- Q8.** Under ideal conditions, the speedup obtained from a k-stage instruction pipeline approaches
(A) 2k (B) $\log k$ (C) k^2 (D) k
- Q9.** A circular queue implemented in an array of size n, keeping one slot always empty to distinguish full from empty, can hold at most
(A) $n - 1$ elements (B) n elements (C) $n/2$ elements (D) $n + 1$ elements
- Q10.** In the array representation of a binary heap with the root at index 1, the children of the node at index i are at indices
(A) $2i$ and $2i + 1$ (B) $i - 1$ and $i + 1$ (C) $i/2$ and $i/2 + 1$ (D) $2i - 1$ and $2i$
- Q11.** Depth-first search of a graph is naturally implemented using
(A) a queue (B) a hash table (C) a stack (D) a priority queue
- Q12.** Matrix chain multiplication is solved optimally using
(A) dynamic programming (B) branch and bound (C) greedy method (D) divide and conquer
- Q13.** The time complexity of building a binary heap from an unsorted array of n elements using the bottom-up procedure is
(A) $O(n \log n)$ (B) $O(\log n)$ (C) $O(n)$ (D) $O(n^2)$
- Q14.** Binary search cannot be applied efficiently to a sorted singly linked list because
(A) recursion is not allowed (B) accessing the middle element takes $O(n)$ time (C) the list is unsorted (D) comparisons are costlier
- Q15.** A context-free grammar is said to be ambiguous if
(A) it generates an infinite language (B) some string has more than one parse tree (C) it is left recursive (D) it has useless productions
- Q16.** The halting problem for Turing machines is
(A) undecidable but recursively enumerable (B) not recursively enumerable (C) context-free (D) decidable
- Q17.** Moving a computation that does not change across iterations out of a loop is called
(A) dead code elimination (B) loop-invariant code motion (C) strength reduction (D) constant folding
- Q18.** The symbol table in a compiler is
(A) created only during code generation (B) built only during lexical analysis (C) discarded after parsing (D) used by almost all phases of the compiler
- Q19.** A counting semaphore initialized to 1 and used with wait/signal around a critical section behaves as
(A) a binary semaphore / mutex (B) a monitor (C) a condition variable (D) a spin lock only
- Q20.** External fragmentation is a problem primarily in
(A) paging (B) demand paging (C) TLB management (D) variable-partition contiguous allocation
- Q21.** Which CPU scheduling algorithm gives the minimum average waiting time for a given set of processes arriving together?
(A) FCFS (B) Priority (C) Shortest Job First (D) Round Robin
- Q22.** In SQL, the clause used to filter groups produced by GROUP BY is
(A) WHERE (B) DISTINCT (C) ORDER BY (D) HAVING
- Q23.** A decomposition of R into R1 and R2 is lossless if
(A) both are in BCNF (B) the common attributes form a superkey of at least one of R1 or R2 (C) $R1 \cap R2$ is empty (D) $R1 \cup R2 \neq R$
- Q24.** The correct sequence of segments in the TCP connection establishment handshake is
(A) SYN, FIN, ACK (B) SYN, SYN+ACK, ACK (C) SYN, ACK, SYN (D) ACK, SYN, SYN+ACK
- Q25.** The default subnet mask of a Class B IPv4 network is
(A) 255.255.0.0 (B) 255.255.255.255 (C) 255.0.0.0 (D) 255.255.255.0
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SHEET 5

- Q1.** A tree has 10 edges. The number of vertices in the tree is
(A) 10 (B) 9 (C) 12 (D) 11
- Q2.** The relation “congruent modulo 5” defined on the set of integers is
(A) not transitive (B) antisymmetric but not reflexive (C) a partial order only (D) an equivalence relation
- Q3.** Simpson’s 1/3 rule for numerical integration requires the number of subintervals to be
(A) any positive integer (B) even (C) prime (D) odd
- Q4.** The dual of the Boolean expression $A + (B \cdot C)$ is
(A) $A' + (B' \cdot C')$ (B) $A \cdot (B + C)$ (C) $A \cdot (B \cdot C)$ (D) $A + (B + C)$
- Q5.** The race-around condition occurs in a
(A) T flip-flop with $T = 0$ (B) SR latch (C) D flip-flop (D) level-triggered JK flip-flop with $J = K = 1$
- Q6.** The interrupt vector table contains
(A) starting addresses of interrupt service routines (B) opcodes of interrupt instructions (C) only interrupt priorities (D) interrupt data
- Q7.** A processor supports 64 distinct instructions. The minimum number of bits required for the opcode field is
(A) 8 (B) 5 (C) 7 (D) 6
- Q8.** The fastest storage element in the memory hierarchy is
(A) L1 cache (B) solid-state disk (C) CPU registers (D) main memory
- Q9.** A recursive function written without a base case will most likely cause
(A) heap fragmentation only (B) a run-time stack overflow (C) an infinite loop using no extra memory (D) a compile-time error
- Q10.** In a full binary tree in which every internal node has exactly 2 children, if there are 10 leaf nodes then the total number of nodes is
(A) 18 (B) 19 (C) 21 (D) 20
- Q11.** The most space-efficient way to store a sparse matrix is
(A) a triplet or linked representation of non-zero elements (B) a stack (C) a two-dimensional array (D) a circular queue
- Q12.** Which of the following functions grows the fastest asymptotically?
(A) $n!$ (B) $n \log n$ (C) 2^n (D) n^2
- Q13.** The fractional knapsack problem is optimally solved by
(A) branch and bound (B) dynamic programming (C) backtracking (D) the greedy method
- Q14.** The time complexity of the Bellman-Ford algorithm on a graph with V vertices and E edges is
(A) $O(E \log V)$ (B) $O(V^2)$ (C) $O(VE)$ (D) $O(V + E)$
- Q15.** Regarding the classes of languages accepted by DFA and NFA
(A) NFAs accept strictly more languages (B) both accept exactly the same class (C) the two classes are incomparable (D) DFAs accept strictly more languages
- Q16.** Context-free languages are closed under
(A) all of the above (B) intersection (C) complementation (D) union
- Q17.** Handle pruning is a technique associated with
(A) semantic analysis (B) lexical analysis (C) top-down parsing (D) bottom-up parsing
- Q18.** An activation record created on the run-time stack typically contains
(A) the complete symbol table (B) local variables, parameters and the return address (C) only global variables (D) generated machine code
- Q19.** A transition of a process from the running state to the ready state is caused by
(A) a page fault (B) process termination (C) an I/O request (D) expiry of the time quantum
- Q20.** For the reference string 1, 2, 3, 4, 1, 2, 5 with 3 frames and FIFO replacement, the number of page faults is
(A) 6 (B) 4 (C) 5 (D) 7
- Q21.** In which threading model does the blocking of one thread NOT block the entire process?
(A) Both behave identically (B) Neither model (C) Pure user-level threads (D) Kernel-level threads
- Q22.** Which ACID property guarantees that the effects of a committed transaction survive subsequent system failures?
(A) Atomicity (B) Consistency (C) Isolation (D) Durability
- Q23.** If relation R has 5 tuples and relation S has 4 tuples, the number of tuples in $R \times S$ is
(A) 4 (B) 9 (C) 20 (D) 5
- Q24.** The well-known default port number for HTTP is
(A) 25 (B) 21 (C) 80 (D) 443
- Q25.** Distance vector routing suffers from which problem?
(A) Silly window syndrome (B) Hidden terminal (C) Count-to-infinity (D) Flooding storm
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SHEET 6

- Q1.** The number of elements in the power set of a set with 5 elements is
(A) 25 (B) 64 (C) 32 (D) 10
- Q2.** For the system $AX = B$ where A is square with $\det(A) = 0$ and $B \neq 0$, the system has
(A) either no solution or infinitely many solutions (B) a unique solution (C) exactly two solutions (D) always infinitely many solutions
- Q3.** Two fair dice are rolled. The probability that the sum of the numbers is 7 is
(A) $1/12$ (B) $5/36$ (C) $1/6$ (D) $1/9$
- Q4.** The number of possible minterms for a Boolean function of 4 variables is
(A) 16 (B) 32 (C) 4 (D) 8
- Q5.** The number of output lines in a 3-to-8 decoder is
(A) 3 (B) 6 (C) 16 (D) 8
- Q6.** The correct order of operations in a basic instruction cycle is
(A) decode, fetch, execute (B) fetch, decode, execute (C) execute, fetch, decode (D) fetch, execute, decode
- Q7.** In a write-back cache
(A) every write updates main memory immediately (B) a dirty bit is not required (C) writes bypass the cache entirely (D) main memory is updated only when the block is replaced
- Q8.** Virtual memory in modern operating systems is normally implemented using
(A) contiguous allocation (B) demand paging (C) swapping of entire processes only (D) overlays only
- Q9.** To achieve $O(1)$ enqueue and $O(1)$ dequeue in a linked-list-based queue, the implementation must maintain
(A) only a head pointer (B) both head and tail pointers (C) a doubly linked list, necessarily (D) only a tail pointer
- Q10.** Given a pointer to a node in a doubly linked list, the time to delete that node is
(A) $O(n)$ (B) $O(\log n)$ (C) $O(1)$ (D) $O(n^2)$
- Q11.** The postorder traversal of an expression tree produces the expression in
(A) infix form (B) postfix form (C) level order (D) prefix form
- Q12.** The maximum number of swaps performed by selection sort on n elements is
(A) n^2 (B) $n - 1$ (C) $n^2/2$ (D) $n \log n$
- Q13.** The solution of the recurrence $T(n) = T(n/2) + 1$ is
(A) $\Theta(\log n)$ (B) $\Theta(1)$ (C) $\Theta(n \log n)$ (D) $\Theta(n)$
- Q14.** A topological ordering of vertices exists for
(A) directed acyclic graphs only (B) directed graphs with cycles (C) any undirected graph (D) any graph
- Q15.** The language $L = \{w \in \{a, b\}^* \mid w \text{ has an equal number of } a\text{'s and } b\text{'s}\}$ is
(A) context-free but not regular (B) finite (C) regular (D) not context-free
- Q16.** Deterministic pushdown automata accept
(A) all context-sensitive languages (B) a proper subset of the context-free languages (C) all context-free languages (D) only regular languages
- Q17.** Peephole optimization is performed on
(A) the symbol table (B) a small sliding window of target code (C) the source program (D) the parse tree
- Q18.** YACC is a tool that generates a
(A) code optimizer (B) lexical analyzer (C) parser (D) assembler
- Q19.** The optimal (OPT/MIN) page replacement algorithm
(A) behaves identically to FIFO (B) suffers from Belady's anomaly (C) is easily implementable (D) gives the lowest possible fault rate but is not implementable in practice
- Q20.** The convoy effect is associated with which scheduling algorithm?
(A) FCFS (B) Round Robin (C) Multilevel feedback queue (D) SJF
- Q21.** In UNIX/Linux, the system call used to create a new process is
(A) `pipe()` (B) `exec()` (C) `wait()` (D) `fork()`
- Q22.** Referential integrity in a relational database is enforced through
(A) a unique constraint (B) a primary key (C) a foreign key (D) a check constraint
- Q23.** A primary index built on the ordering key field of an ordered data file is normally
(A) a secondary index (B) dense (C) always multilevel (D) sparse
- Q24.** Cyclic Redundancy Check (CRC) is used for
(A) flow control (B) error detection (C) encryption (D) routing
- Q25.** Network Address Translation (NAT) operates at which layer?
(A) Session (B) Data link (C) Network (D) Application
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SHEET 7

- Q1.** The number of distinct labelled spanning trees of the complete graph K_4 is
(A) 16 (B) 64 (C) 8 (D) 4
- Q2.** The eigenvalues of the matrix $\begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$ are
(A) 2 and 3 (B) 1 and 2 (C) 1 and 3 (D) 0 and 3
- Q3.** A bag contains 3 red and 2 blue balls. Two balls are drawn one after another without replacement. The probability that both are red is
(A) $9/25$ (B) $1/2$ (C) $3/10$ (D) $2/5$
- Q4.** The number of 2-input NAND gates required to realize a NOT gate is
(A) 1 (B) 2 (C) 4 (D) 3
- Q5.** The Gray code equivalent of the binary number 1010 is
(A) 1101 (B) 1010 (C) 0101 (D) 1111
- Q6.** The throughput of a linear instruction pipeline is determined by
(A) the slowest stage (B) the number of stages alone (C) the fastest stage (D) the average stage delay
- Q7.** A 2-way set-associative cache contains 128 blocks in total. The number of sets is
(A) 16 (B) 64 (C) 256 (D) 128
- Q8.** A distinguishing feature of a synchronous bus is
(A) the absence of control lines (B) fully variable transfer timing (C) handshaking signals for every transfer (D) a common clock shared by all devices
- Q9.** In C, when an array is passed as an argument to a function, what is actually passed is
(A) a copy of the entire array (B) a reference to the whole array object (C) a pointer to its first element (D) the array size only
- Q10.** The minimum number of nodes in an AVL tree of height 3 (root at height 0) is
(A) 6 (B) 8 (C) 7 (D) 15
- Q11.** The most efficient data structure for implementing a priority queue supporting $O(\log n)$ insert and extract-min is
(A) a stack (B) a binary heap (C) a sorted array (D) an unsorted linked list
- Q12.** The average-case time complexity of quicksort is
(A) $O(n)$ (B) $O(n^2)$ (C) $O(n \log n)$ (D) $O(\log n)$
- Q13.** The dynamic programming solution for the longest common subsequence of strings of lengths m and n takes
(A) $O(mn)$ time (B) $O(n \log n)$ time (C) $O(m + n)$ time (D) $O(2^n)$ time
- Q14.** Prim's algorithm implemented with an adjacency matrix and no priority queue runs in
(A) $O(V^3)$ (B) $O(V + E)$ (C) $O(E \log V)$ (D) $O(V^2)$
- Q15.** The regular expression denoting all strings of even length over the alphabet $\{a\}$ is
(A) a^* (B) $(a + aa)^*$ (C) $(aa)^*$ (D) $a(aa)^*$
- Q16.** The grammar $S \rightarrow aSb \mid \epsilon$ generates the language
(A) all strings over $\{a, b\}$ (B) $\{a^n b^n \mid n \geq 0\}$ (C) $\{a^n b^m \mid n, m \geq 0\}$ (D) $(ab)^*$
- Q17.** A recursive descent parser is an example of a
(A) LR parser (B) bottom-up parser (C) operator precedence parser (D) top-down parser
- Q18.** A basic block in intermediate code is a sequence of instructions with
(A) exactly one instruction (B) multiple entry points (C) at least one loop (D) a single entry point and a single exit point
- Q19.** Mutual exclusion implemented using the TestAndSet instruction is an example of
(A) interrupt disabling only (B) a purely software solution (C) a compiler directive (D) a hardware-supported solution involving busy waiting
- Q20.** Internal fragmentation is an inherent problem of
(A) neither (B) both equally (C) paging (D) pure segmentation
- Q21.** The working set model is used primarily to
(A) improve cache hit ratio (B) allocate disk blocks (C) speed up CPU scheduling (D) prevent thrashing
- Q22.** Which of the following is a DDL command?
(A) DELETE (B) CREATE (C) SELECT (D) UPDATE
- Q23.** A schedule is conflict serializable if and only if its precedence (serializability) graph
(A) is acyclic (B) contains a cycle (C) is a complete graph (D) is bipartite
- Q24.** IEEE 802.11 wireless LANs use which medium access method?
(A) CSMA/CA (B) Pure ALOHA (C) Token ring (D) CSMA/CD
- Q25.** The number of $/27$ subnets that can be carved out of a single $/24$ network is
(A) 4 (B) 8 (C) 32 (D) 16
-

SHEET 8

- Q1.** The total number of binary relations that can be defined on a set of 3 elements is
(A) 27 (B) 64 (C) 512 (D) 9
- Q2.** A square matrix A is singular if
(A) A is symmetric (B) $\det(A) = 0$ (C) $\det(A) \neq 0$ (D) A is orthogonal
- Q3.** The standard deviation of the data 2, 4, 4, 4, 5, 5, 7, 9 is
(A) 1 (B) 4 (C) 2 (D) 2.5
- Q4.** The function $F(A, B) = A \oplus B$ expressed as a sum of minterms is
(A) $\Sigma m(0, 1)$ (B) $\Sigma m(2, 3)$ (C) $\Sigma m(0, 3)$ (D) $\Sigma m(1, 2)$
- Q5.** The number of flip-flops required to build a 4-bit shift register is
(A) 2 (B) 4 (C) 8 (D) 16
- Q6.** The program counter is normally incremented during the
(A) fetch phase (B) execute phase of an ALU operation (C) operand decode phase (D) interrupt phase only
- Q7.** Memory interleaving is used mainly to improve
(A) effective memory bandwidth (B) memory reliability (C) memory capacity (D) cache size
- Q8.** Which of the following is a characteristic feature of RISC architecture?
(A) Complex microprogrammed control for most instructions (B) A very large number of addressing modes (C) Variable-length instructions (D) Fixed-length instructions
- Q9.** Elements 1, 2, 3 are pushed onto a stack in that order, with pops interleaved arbitrarily. Which output sequence is NOT possible?
(A) 1, 3, 2 (B) 3, 2, 1 (C) 3, 1, 2 (D) 2, 3, 1
- Q10.** A binary tree with n nodes has how many NULL child pointers?
(A) $n - 1$ (B) n (C) $2n$ (D) $n + 1$
- Q11.** In a threaded binary tree, NULL pointers are used to
(A) store additional data (B) point to the in-order predecessor or successor (C) reduce the height of the tree (D) balance the tree
- Q12.** The best-case time complexity of insertion sort is
(A) $O(n^2)$ (B) $O(1)$ (C) $O(n)$ (D) $O(n \log n)$
- Q13.** Which sorting algorithm runs in $O(n)$ time when the input consists of integers drawn from a small fixed range?
(A) Heapsort (B) Counting sort (C) Quicksort (D) Merge sort
- Q14.** The Floyd-Warshall all-pairs shortest path algorithm has time complexity
(A) $O(V^3)$ (B) $O(V^2)$ (C) $O(VE)$ (D) $O(E \log V)$
- Q15.** The number of states in the minimal DFA for the regular expression $(0 + 1)^*11$ is
(A) 2 (B) 3 (C) 4 (D) 5
- Q16.** A language accepted by a Turing machine that halts on every input is
(A) recursively enumerable but not recursive (B) necessarily context-free (C) recursive (decidable) (D) not recursively enumerable
- Q17.** FIRST and FOLLOW sets are computed for the purpose of
(A) constructing an LL(1) parsing table (B) register allocation (C) lexical analysis (D) peephole optimization
- Q18.** Register allocation and assignment is performed during
(A) lexical analysis (B) parsing (C) semantic analysis (D) code generation
- Q19.** The problem of priority inversion in real-time systems is addressed by
(A) aging (B) round robin scheduling (C) FCFS scheduling (D) priority inheritance
- Q20.** Contiguous allocation of disk files provides
(A) fast direct access but suffers from external fragmentation (B) no fragmentation at all (C) mandatory use of an index block (D) slow direct access
- Q21.** The standard semaphore solution to the bounded-buffer producer-consumer problem uses how many semaphores?
(A) 2 (B) 4 (C) 3 (D) 1
- Q22.** A trigger in a relational database is
(A) a virtual table (B) an index structure (C) a constraint checked manually (D) a procedure executed automatically in response to a specified event
- Q23.** A view in SQL is
(A) a virtual table derived from a query (B) an index (C) a physically stored table (D) a backup file
- Q24.** Which transport layer protocol is connectionless?
(A) FTP (B) TCP (C) UDP (D) SCTP
- Q25.** The TTL field in an IPv4 header prevents
(A) fragmentation (B) congestion (C) duplicate acknowledgements (D) packets from circulating indefinitely
-

SHEET 9

- Q1.** The number of ways of selecting 3 objects out of 8 distinct objects is
(A) 24 (B) 112 (C) 336 (D) 56
- Q2.** The sum of all eigenvalues of a square matrix is equal to its
(A) norm (B) determinant (C) rank (D) trace
- Q3.** For the recurrence $a_n = a_{n-1} + 2$ with $a_1 = 1$, the value of a_{10} is
(A) 21 (B) 19 (C) 17 (D) 20
- Q4.** The expression $A + A'B$ simplifies to
(A) AB (B) A (C) $A + B$ (D) B
- Q5.** The SUM output of a half adder with inputs A and B is
(A) $A \oplus B$ (B) $A + B$ (C) $A \cdot B$ (D) $(A \cdot B)'$
- Q6.** The number of address lines required to address 1 MB of byte-addressable memory is
(A) 18 (B) 20 (C) 24 (D) 16
- Q7.** A cache has an access time of 10 ns and main memory 100 ns. The hit ratio is 0.95 and on a miss only main memory is accessed. The average memory access time is
(A) 10 ns (B) 19 ns (C) 14.5 ns (D) 55 ns
- Q8.** Booth's algorithm is used for
(A) binary division (B) floating point addition (C) multiplication of signed binary numbers (D) normalization of mantissa
- Q9.** In a C program, memory obtained through malloc() is allocated from the
(A) code segment (B) stack (C) data segment (D) heap
- Q10.** The time to search for a key in a height-balanced binary search tree with n nodes is
(A) $O(n \log n)$ (B) $O(\log n)$ (C) $O(1)$ (D) $O(n)$
- Q11.** The level-order traversal of a binary tree is implemented using
(A) a queue (B) a max-heap (C) a BST (D) a stack
- Q12.** Which of the following employ the divide-and-conquer strategy?
(A) All of merge sort, binary search and quicksort (B) Merge sort only (C) Binary search only (D) Quicksort only
- Q13.** The greedy approach does NOT yield an optimal solution for
(A) the fractional knapsack problem (B) Huffman coding (C) the 0/1 knapsack problem (D) minimum spanning tree
- Q14.** Merging two sorted arrays of sizes m and n requires, in the worst case,
(A) $m + n - 1$ comparisons (B) $\max(m, n)$ comparisons (C) mn comparisons (D) $m + n$ comparisons
- Q15.** In the Chomsky hierarchy, a Type-3 grammar generates
(A) recursively enumerable languages (B) context-free languages (C) regular languages (D) context-sensitive languages
- Q16.** The intersection of a context-free language with a regular language is always
(A) not context-free (B) regular (C) context-free (D) finite
- Q17.** Left factoring of a grammar is performed to make it suitable for
(A) peephole optimization (B) lexical analysis (C) LR parsing (D) predictive (LL(1)) parsing
- Q18.** The DAG representation of a basic block is useful for
(A) tokenizing (B) register spilling only (C) common sub-expression elimination (D) parsing
- Q19.** Four processes arrive at time 0 with CPU bursts 6, 8, 7 and 3 units. Under SJF scheduling the average waiting time is
(A) 8 units (B) 6.5 units (C) 7 units (D) 10.25 units
- Q20.** The Translation Lookaside Buffer (TLB) is
(A) part of secondary storage (B) a page replacement policy (C) a cache of recently used page table entries (D) a file allocation index
- Q21.** Spinlocks are the preferred synchronization mechanism when
(A) critical sections are very long (B) critical sections are very short and multiple processors are available (C) processes never contend for the lock (D) there is a single CPU and long critical sections
- Q22.** For R(A, B, C) with the FD set $\{A \rightarrow B, B \rightarrow C\}$, the candidate key is
(A) A (B) BC (C) C (D) B
- Q23.** Deadlock prevention in database concurrency control is achieved by schemes such as
(A) checkpointing (B) shadow paging (C) wait-die and wound-wait (D) two-phase commit
- Q24.** The layer responsible for reliable end-to-end (process-to-process) delivery is the
(A) data link layer (B) transport layer (C) session layer (D) network layer
- Q25.** Which DNS resource record maps a domain name to an IPv4 address?
(A) CNAME (B) PTR (C) A (D) MX
-

SHEET 10

- Q1.** If $|A| = 3$ and $|B| = 4$, the number of functions from A to B is
(A) 81 (B) 64 (C) 12 (D) 24
- Q2.** In a partially ordered set, the least upper bound of a pair of elements is called the
(A) complement (B) minimal element (C) supremum (D) infimum
- Q3.** For events A and B with $P(B) > 0$, the conditional probability $P(A | B)$ equals
(A) $P(A \cup B)/P(B)$ (B) $P(A \cap B)/P(A)$ (C) $P(A) \cdot P(B)$ (D) $P(A \cap B)/P(B)$
- Q4.** The output of a 2-input XNOR gate is 1 when
(A) both inputs are 0 only (B) both inputs are 1 only (C) the inputs differ (D) the inputs are equal
- Q5.** The number of distinct states in a MOD-16 counter is
(A) 15 (B) 16 (C) 8 (D) 4
- Q6.** The instruction `MOV R1, #5` uses which addressing mode?
(A) Indirect (B) Immediate (C) Direct (D) Indexed
- Q7.** Translation of a logical address into a physical address at run time is performed by the
(A) compiler (B) DMA controller (C) ALU (D) memory management unit
- Q8.** In a system with a two-level page table and no TLB hit, the number of memory accesses needed to read one data word is
(A) 2 (B) 1 (C) 3 (D) 4
- Q9.** In C, the value returned by `strlen("hello")` is
(A) 5 (B) 4 (C) undefined (D) 6
- Q10.** The worst-case height of a binary search tree containing n nodes is
(A) $n/2$ (B) $\log_2 n$ (C) $\sqrt{n}$ (D) $n - 1$
- Q11.** Linear probing as a collision resolution technique suffers from
(A) secondary clustering (B) primary clustering (C) no clustering at all (D) chaining overhead
- Q12.** Which algorithm selects the kth smallest element of an unsorted array in $O(n)$ worst-case time?
(A) Quickselect with a random pivot (B) Median-of-medians selection (C) Heapsort (D) Merge sort
- Q13.** Strassen's algorithm multiplies two $n \times n$ matrices in approximately
(A) $O(n^2)$ (B) $O(n^3)$ (C) $O(n \log n)$ (D) $O(n^{2.81})$
- Q14.** Huffman coding is
(A) a divide-and-conquer algorithm (B) a greedy but sub-optimal algorithm (C) a greedy algorithm that produces an optimal prefix code (D) a dynamic programming algorithm producing fixed-length codes
- Q15.** Which of the following containments is correct?
(A) $CFL \subset Regular$ (B) $Regular \subset CFL \subset Recursive$ (C) $Regular = CFL$ (D) $Recursive \subset CFL$
- Q16.** The problem of deciding whether the language generated by a given context-free grammar is empty is
(A) undecidable (B) not recursively enumerable (C) decidable (D) NP-complete but undecidable
- Q17.** An L-attributed syntax-directed definition permits inherited attributes to be computed from
(A) no other nodes (B) children only (C) the parent and left siblings (D) right siblings only
- Q18.** Which of the following is a machine-independent optimization?
(A) Common sub-expression elimination (B) Register allocation (C) Peephole optimization (D) Instruction selection
- Q19.** As the time quantum in Round Robin scheduling is made very small,
(A) starvation of long processes increases (B) context switch overhead increases (C) it behaves like SJF (D) context switch overhead decreases
- Q20.** Which disk scheduling algorithm can cause starvation of requests located far from the current head position?
(A) SSTF (B) SCAN (C) C-SCAN (D) FCFS
- Q21.** A race condition occurs when
(A) CPU utilization reaches 100% (B) a process waits indefinitely for a resource (C) processes never share data (D) the final result depends on the relative order of accesses to shared data
- Q22.** In SQL, `COUNT(column_name)`
(A) counts NULL values as well (B) is invalid syntax (C) ignores NULL values (D) is always equal to `COUNT(*)`
- Q23.** The minimum possible number of tuples in the natural join of two non-empty relations is
(A) 0 (B) the size of the smaller relation (C) 1 (D) the product of the two sizes
- Q24.** The Address Resolution Protocol (ARP) maps
(A) a MAC address to an IP address (B) a domain name to an IP address (C) a port number to a process (D) an IP address to a MAC address
- Q25.** A sliding window protocol with a sender window size of 1 reduces to
(A) Go-Back-N (B) Stop-and-Wait (C) piggybacked sliding window (D) Selective Repeat
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ANSWER KEYS

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